Pearson Edexcel Level 1/Level 2 – style practice paper

GCSE (9–1) Mathematics

1MA1/1H & 2H – Higher Tier

Topic 3: Ratio, proportion and rates of change

Centre Number		**Candidate Number**	
Surname		**Other names**	

Practice questions in the style of Pearson Edexcel GCSE Mathematics (1MA1). Every question is original; the layout, command words, mark allocations and mark scheme (M, A, B marks) follow the real exam format. Calculator rules are given in the instructions below.

Time allowed	About 100 minutes (roughly 1 minute per mark)
Instructions	• Use **black** ink or ball-point pen. • Answer **all** questions. • Answer the questions in the spaces provided – there may be more space than you need. • Calculators must **not** be used in Section A. Calculators may be used in Section B. • Diagrams are **NOT** accurately drawn, unless otherwise indicated. • You must **show all your working**. • **Unless stated, give non-exact answers to 3 significant figures.**
Information	• The total mark for this paper is **99**. • The marks for **each** question are shown in brackets – use this as a guide as to how much time to spend on each question. • Questions labelled with an asterisk (*) are ones where the quality of your written communication will be assessed.
Advice	• Read each question carefully before you start to answer it. • Keep an eye on the time. • Try to answer every question. • Check your answers if you have time at the end.

Name: _____ Date: _____ Mark: _____ / 99

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Section A: Non-calculator

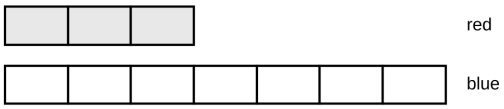
You must not use a calculator for questions 1 to 14.

1 Share £240 in the ratio 3 : 5 : 4 **(3)**

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(Total for Question 1 is 3 marks)

2 In a bag, the ratio of red counters to blue counters is 3 : 7. There are 20 more blue counters than red counters.



Work out the total number of counters in the bag. **(3)**

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(Total for Question 2 is 3 marks)

3 A shop sells apples in two packs. Pack A: 6 apples for £2.10. Pack B: 10 apples for £3.40. Which pack is the better value? You must show your working. **(3)**

.....

(Total for Question 3 is 3 marks)

4 (a) A recipe for 8 people uses 600 g of flour. Work out how much flour is needed for 14 people. **(2)**

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(b) The map scale is 1 : 25 000. A path is 6.4 cm long on the map. Work out the real length of the path in kilometres. **(3)**

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(Total for Question 4 is 5 marks)

5 (a) A car travels 150 km in 2 hours 30 minutes. Work out its average speed. **(2)**

.....

(b) Change 72 km/h into metres per second. **(2)**

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(c) Change 5 m/s into kilometres per hour. **(2)**

.....

(Total for Question 5 is 6 marks)

6 (a) A metal block has mass 540 g and volume 200 cm³. Work out the density of the metal. **(2)**

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(b) Another block of the same metal has a mass of 1.35 kg. Work out its volume in cm³. **(3)**

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(c) A force of 600 N acts on an area of 0.25 m². Work out the pressure. **(2)**

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(Total for Question 6 is 7 marks)

7 (a) y is directly proportional to x^2. When $x = 5$, $y = 50$. Find y when $x = 8$. **(3)**

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(b) Find the positive value of x when $y = 18$. **(2)**

.....

(c) p is inversely proportional to q. When $q = 4$, $p = 6$. Find p when $q = 3$. **(3)**

.....

(d) m is inversely proportional to the square of n. When $n = 4$, $m = 3$. Find n when $m = 12$. **(3)**

.....

(Total for Question 7 is 11 marks)

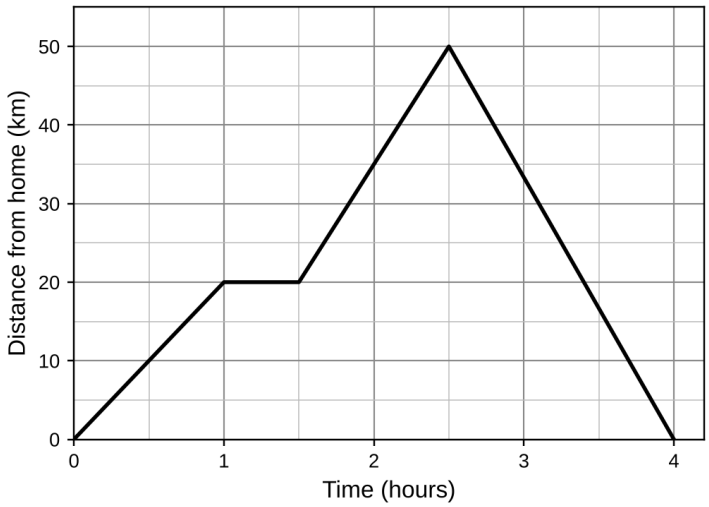
8 The ratio of a to b is 3 : 4. The ratio of b to c is 2 : 5.

Find the ratio $a : b : c$. **(3)**

.....

(Total for Question 8 is 3 marks)

9 The graph shows Tom's journey on his bike from home to a lake and back again.



(a) Work out his speed during the first hour. **(2)**

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(b) For how long did Tom stop? **(1)**

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(c) Work out Tom's speed on the return journey. **(3)**

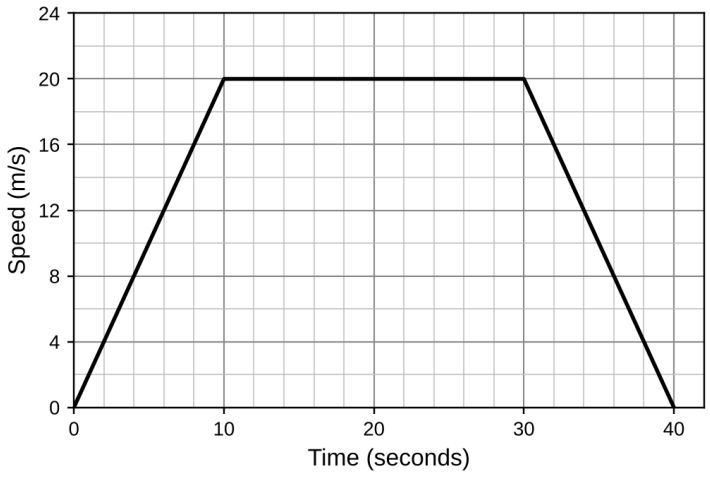
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(d) Work out Tom's average speed for the whole journey, including the stop. **(3)**

.....

(Total for Question 9 is 9 marks)

10 The graph shows the speed of a train during part of its journey.



(a) Work out the acceleration of the train in the first 10 seconds. **(2)**

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(b) Work out the distance the train travels in the 40 seconds. **(3)**

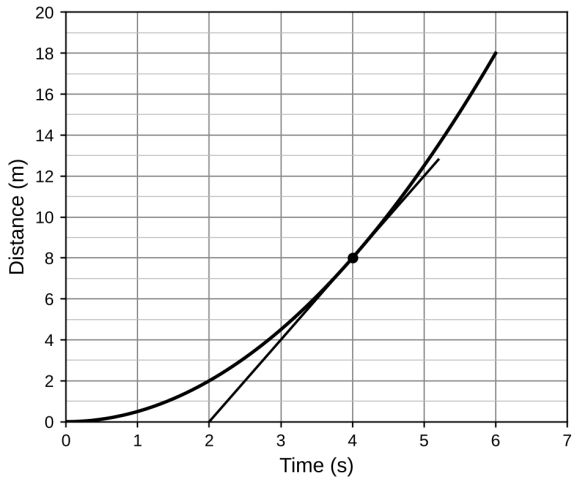
.....

(c) Work out the average speed of the train for the 40 seconds. **(2)**

.....

(Total for Question 10 is 7 marks)

11 The graph shows the distance travelled by a runner during the first 6 seconds of a race. A tangent has been drawn at $t = 4$.



Use the tangent to estimate the speed of the runner at $t = 4$. **(3)**

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(Total for Question 11 is 3 marks)

12 A bus travels 60 km at an average speed of 40 km/h. It then travels 60 km at an average speed of 60 km/h. Work out the average speed for the whole journey. **(4)**

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(Total for Question 12 is 4 marks)

13 The time, T hours, taken to fill a reservoir is inversely proportional to the number of pumps, n. When 4 pumps are used it takes 6 hours. How many pumps are needed to fill the reservoir in 4 hours? **(3)**

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(Total for Question 13 is 3 marks)

14 The sides of two rectangles are in the ratio 2 : 3. The area of the smaller rectangle is 48 cm². Work out the area of the larger rectangle. **(3)**

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(Total for Question 14 is 3 marks)

Section B: Calculator

A calculator may be used for questions 15 to 22.

15 Water flows into a tank at a rate of 12 litres per minute. The tank is a cuboid measuring 1.5 m by 1 m by 1 m. How long will it take to fill the empty tank? Give your answer in hours and minutes. **(4)**

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(Total for Question 15 is 4 marks)

16 £1 = €1.17. A hotel in France costs €84 per night. Another hotel in England costs £70 per night.

Which hotel is cheaper? You must show your working. **(3)**

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(Total for Question 16 is 3 marks)

17 A pipe fills a pool at 15 litres per minute. A second pipe fills it at 10 litres per minute. The pool holds 90 litres. Work out how long both pipes together take to fill the pool. Give your answer in minutes and seconds. **(3)**

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(Total for Question 17 is 3 marks)

18 Anya drives 87 miles in 1 hour 45 minutes. Work out her average speed in kilometres per hour. Use 5 miles = 8 kilometres. **(4)**

.....

(Total for Question 18 is 4 marks)

19 The population of a town is 25 000. It increases at a rate of 3.5% per year. The mayor says, "In 20 years the population will have more than doubled."

Is the mayor correct? You must show your working. **(3)**

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(Total for Question 19 is 3 marks)

20 The density of gold is 19.3 g/cm³. A gold bar is a cuboid 10 cm by 5 cm by 2 cm. Work out the mass of the bar in kilograms. **(3)**

.....

(Total for Question 20 is 3 marks)

21 A cyclist rides at 18 km/h. A runner runs at 4.5 m/s.

(a) Who is faster? You must show your working. **(3)**

.....

(b) The runner runs for 2 hours at this speed. How far does she run? Give your answer in kilometres to 1 decimal place. **(3)**

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(Total for Question 21 is 6 marks)

22 Sophie buys a laptop in a sale. The normal price is reduced in the ratio 5 : 4. The sale price is £616.

Work out the normal price of the laptop. **(3)**

£ .....

(Total for Question 22 is 3 marks)

TOTAL FOR PAPER IS 99 MARKS

END

MARK SCHEME – Topic 3: Ratio, proportion and rates of change

Guidance on marking: M marks are method marks, awarded for a correct method (even if the final answer is wrong). **A** marks are accuracy marks, awarded only if the relevant M mark has been gained. **B** marks are independent marks (no method needed). **C** marks are communication marks. **P** marks are process marks (a correct approach to a problem). **ft** = follow through from a previous incorrect answer. **oe** = or equivalent. **cao** = correct answer only. **SC** = special case. If the answer on the answer line differs from the working, the answer line is marked unless the working is clearly the intended answer. Correct answers with no working score full marks unless the question says 'You must show your working'.

Question	Working	Answer	Mark	Notes
1	M1 $3+5+4=12$ and $240\div12=20$ M1 $3\times20,\ 5\times20,\ 4\times20$ A1 £60, £100, £80	£ 60, £ 100, £ 80	3 AO1	
2	M1 $7-3=4$ parts $=20$ M1 1 part $=5$ and 10 parts A1 50	50	3 AO3	
3	M1 $210\div6=35$p or $340\div10=34$p (or apples per £) M1 both unit costs found (or both amounts for same number of apples) A1 Pack B with correct figures (34p < 35p)	Pack B	3 AO3	
4(a)	M1 $600\div8\times14$ or 75 g per person A1 1050	1050 g	2 AO1	
4(b)	M1 $6.4\times25000=160000$ cm M1 $160000\div100\div1000$ A1 1.6	1.6 km	3 AO1	
5(a)	M1 $150\div2.5$ A1 60	60 km/h	2 AO1	
5(b)	M1 $72\times1000\div3600$ A1 20	20 m/s	2 AO1	
5(c)	M1 $5\times3600\div1000$ A1 18	18 km/h	2 AO1	
6(a)	M1 $540\div200$ A1 2.7	2.7 g/cm³	2 AO1	
6(b)	B1 1.35 kg $=1350$ g M1 $1350\div2.7$ A1 500	500 cm³	3 AO2	
6(c)	M1 $600\div0.25$ A1 2400	2400 N/m²	2 AO1	
7(a)	M1 $y=kx^2$ and $50=k\times25$ A1 $k=2$ A1 $y=128$	128	3 AO1	
7(b)	M1 $18=2x^2$ so $x^2=9$ A1 3	3	2 AO1	
7(c)	M1 $p=\frac{k}{q}$ A1 $k=24$ A1 8	8	3 AO1	

Question	Working	Answer	Mark	Notes
7(d)	M1 $m = \frac{k}{n^2}$ and $3 = \frac{k}{16}$, so $k = 48$ M1 $12 = \frac{48}{n^2}$ so $n^2 = 4$ A1 $n = 2$ (positive)	2	3 AO1	
8	M1 make b the same: $3 : 4 = 6 : 8$ and $2 : 5 = 8 : 20$ M1 $a : b : c = 6 : 8 : 20$ A1 $3 : 4 : 10$	3 : 4 : 10	3 AO2	
9(a)	M1 $20 \div 1$ A1 20	20 km/h	2 AO1	
9(b)	B1 30 minutes (0.5 hours)	30 minutes	1 AO1	
9(c)	M1 distance 50 km, time $4 - 2.5 = 1.5$ hours M1 $50 \div 1.5$ A1 33.3 (accept $33\frac{1}{3}$)	33.3 km/h	3 AO1	
9(d)	B1 total distance $= 100$ km M1 $100 \div 4$ A1 25	25 km/h	3 AO3	
10(a)	M1 $20 \div 10$ A1 2	2 m/s²	2 AO1	
10(b)	M1 area of trapezium or sum of areas under the graph M1 $\frac{1}{2}(20 + 40) \times 20$ (oe) A1 600	600 m	3 AO1	
10(c)	M1 $600 \div 40$ A1 15	15 m/s	2 AO1	
11	M1 uses two points on the tangent, e.g. $(2, 0)$ and $(5.2, 12.8)$ M1 $\frac{12.8 - 0}{5.2 - 2}$ A1 4 (accept 3.5 to 4.5)	4 m/s	3 AO3	
12	M1 $60 \div 40 = 1.5$ h M1 $60 \div 60 = 1$ h M1 $120 \div 2.5$ A1 48	48 km/h	4 AO3	Common error: $(40 + 60) \div 2 = 50$ scores 0
13	M1 $T = \frac{k}{n}$ and $6 = \frac{k}{4}$ A1 $k = 24$ A1 6	6	3 AO3	
14	M1 area scale factor $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$ M1 $48 \times \frac{9}{4}$ A1 108	108 cm²	3 AO3	
15	M1 volume $1.5 \times 1 \times 1 = 1.5$ m³ M1 1.5 m$^3 = 1500$ litres M1 $1500 \div 12 = 125$ minutes A1 2 hours 5 minutes	2 hours 5 minutes	4 AO3	

Question	Working	Answer	Mark	Notes
16	M1 $84 \div 1.17$ (= £71.79…) or $70 \times 1.17 = 81.9$ M1 comparison in the same currency A1 England hotel (£70 < £71.79)	England	3 AO3	
17	M1 $15 + 10 = 25$ litres per minute M1 $90 \div 25 = 3.6$ minutes A1 3 minutes 36 seconds	3 minutes 36 seconds	3 AO3	
18	M1 $87 \div 1.75 = 49.7\ldots$ mph M1 $\times \frac{8}{5}$ or $87 \times 1.6 = 139.2$ km M1 $139.2 \div 1.75$ A1 79.5 (km/h)	79.5 km/h	4 AO3	
19	M1 25000×1.035^{20} M1 $= 49744.7\ldots$ C1 $49745 < 50000$ so the mayor is not correct	No (49 745)	3 AO3	
20	M1 volume $= 10 \times 5 \times 2 = 100$ cm³ M1 $100 \times 19.3 = 1930$ g A1 1.93	1.93 kg	3 AO1	
21(a)	M1 $4.5 \times 3.6 = 16.2$ km/h or $18 \div 3.6 = 5$ m/s M1 comparison in same units A1 the cyclist	The cyclist	3 AO3	
21(b)	M1 4.5×7200 M1 $= 32400$ m A1 32.4	32.4 km	3 AO1	
22	M1 $616 \div 4 \times 5$ M1 154×5 A1 770	£ 770	3 AO1	

Where marks are lost in Topic 3

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Ratio:** add the parts first, divide to find one part, then multiply. Check the shares add back to the total.
- ☐ **Unit conversions:** km/h → m/s means ×1000 ÷ 3600 (÷ 3.6). Write the units on every line.
- ☐ **Average speed** = total distance ÷ total time. It is *never* the mean of two speeds when the times differ.
- ☐ **Direct and inverse proportion:** always write the equation with k, find k with the given pair, then use it. Inverse: $y = \frac{k}{x}$.
- ☐ **Graphs:** gradient of a distance–time graph = speed; gradient of a speed–time graph = acceleration; area under a speed–time graph = distance.
- ☐ **Tangent to a curve:** draw the tangent with a ruler, choose two points far apart, use rise ÷ run.
- ☐ **Similar shapes:** area scales by the square of the length scale factor; volume by the cube.
- ☐ **Compound measures:** density = mass ÷ volume, pressure = force ÷ area. Convert units (kg → g, m^3 → cm^3 or litres) before dividing.